12.4 Inner Lane Closure – Multilane Roundabout – Short and Long Duration

Purpose:

This layout shows the typical setup for an inner lane closure within a multilane roundabout.

Traffic Control Persons may not be required because traffic should be able to self-regulate, using the roundabout as if it were a single-lane roundabout.

Standard:

- A Left Lane Closed C-030-3A sign is required in advance of the roundabout, followed by the corresponding Left Lane Closed Ahead C-130-L sign.
- A flashing arrow board (FAB) shall be used inside each taper.
- A Lane Closure Arrow C-053 sign is required on the outside edge of the inner lane.
- Channelizing devices shall be used to isolate the work area and separate the inner and outer lanes.

Guidance:

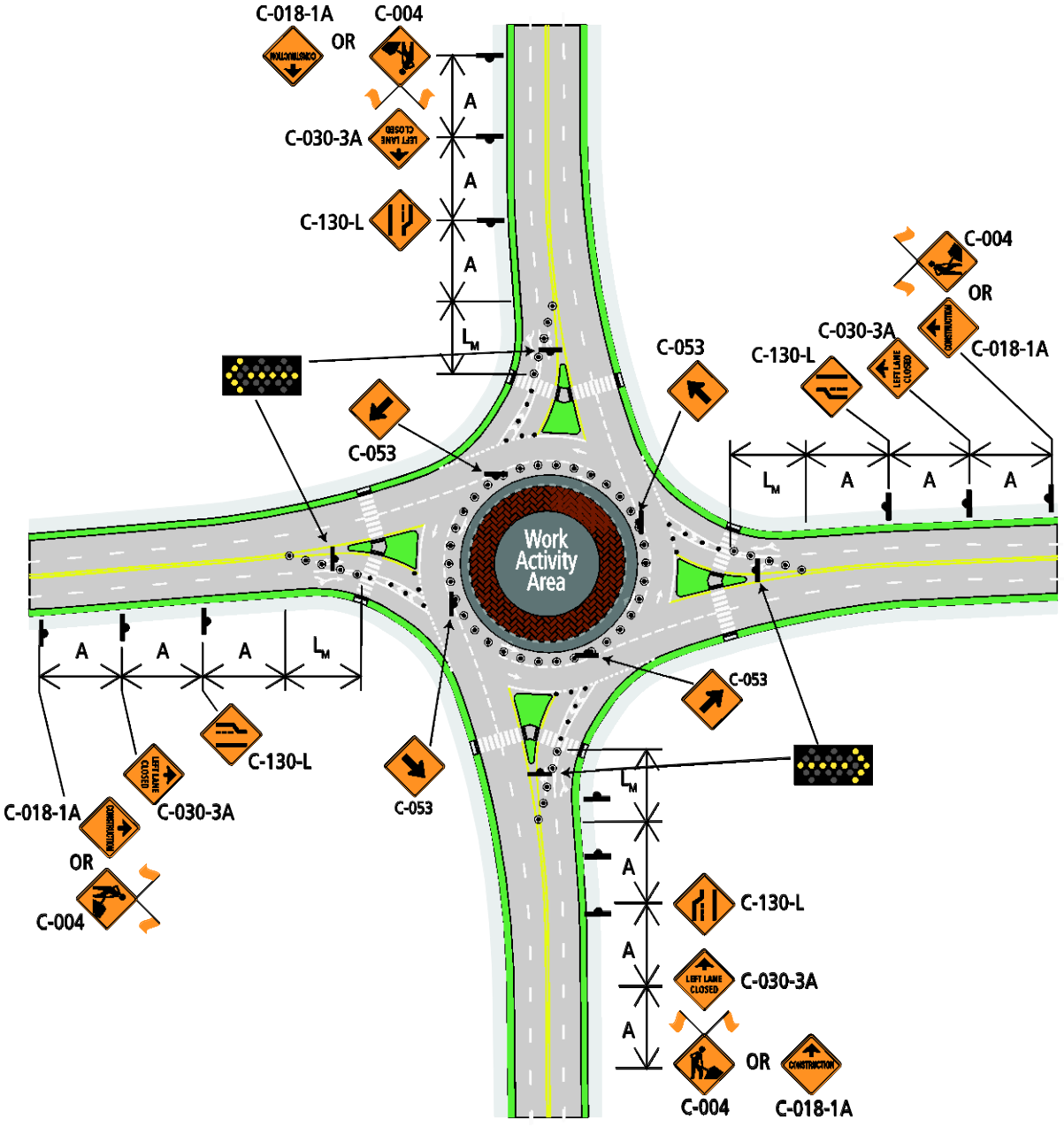
- Existing signs regarding lane use may have to be covered because the roundabout will be operating as a single-lane roundabout.
- Device positioning may have to be adjusted to accommodate long and combination vehicles, which require more room to navigate through roundabouts.

Options:

- Where approach speeds are ≤60 km/h, cones may be used instead of tubular markers.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 12.4: Inner Lane Closure – Multilane Roundabout – Short and Long Duration



12.5 Outer Lane Closure – Multilane Roundabout – Short and Long Duration

Purpose:

This layout shows the typical setup for an outer lane closure within a multilane roundabout.

It is used to isolate the work area and guide traffic through the roundabout in the inner lane while also leaving space for traffic to enter and exit the inner lane.

Traffic Control Persons may not be required because traffic should be able to self-regulate, using the roundabout as if it were a single-lane roundabout.

Standard:

- A Right Lane Closed C-030-4A sign is required in advance of the roundabout, followed by the corresponding Right Lane Closed Ahead C-130-R sign.
- A flashing arrow board (FAB) shall be used inside each taper.
- A Lane Closure Arrow C-053 sign shall be positioned just inside the work area to warn traffic navigating the roundabout that the outer lane is closed.
- Barricades are required on approaches to the work activity area for long-duration work.

Guidance:

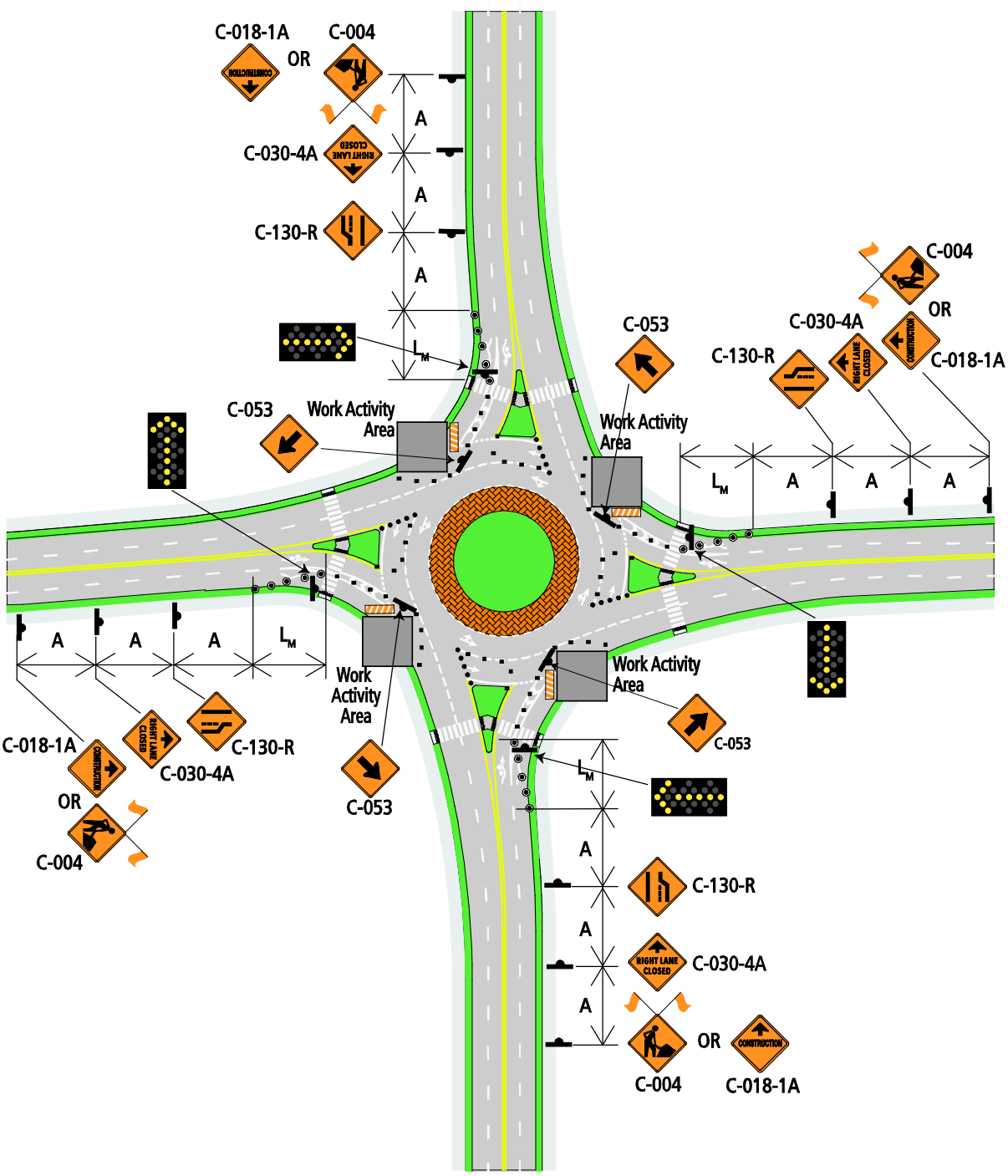
- Device positioning may have to be adjusted to accommodate long and combination vehicles, which require more room to navigate through roundabouts.
- It may be difficult for large vehicles to turn right to exit the roundabout. Additional guidance can be provided by using the Roundabout Right Turn Truck Signs C-121-1 series, which directs drivers to circumnavigate the roundabout so that they re-approach the exit straight-on.
- It may be necessary to detour large trucks from the area during construction.

Options:

- Where approach speeds are ≤60 km/h, cones may be used instead of tubular markers.
- The flashing arrow board (FAB) may be replaced as shown below.

Speed Limit	Workers or Work Vehicles Present	No Workers or Vehicles Present
≤ 60 km/h	Lane Closure Arrow C-053 sign plus 360° flashing light and 4-way flashers.	Lane Closure Arrow C-053 sign plus barricade and Type A yellow warning light.
≥ 70 km/h	No substitution.	

Figure 12.5: Outer Lane Closure – Multilane Roundabout – Short and Long Duration



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PART B – TRAFFIC CONTROL

Section 13: Traffic Control Layouts – Milling, Paving, Seal Coating

Contents

Legend, Table A, and Table B 13-1

13.1 General Information 13-2

13.2 Benkleman Beam and Falling Weight Deflectometer Testing 13-5

13.3 Advance Warning for Paving Work 13-6

13.4 Paving – Work in Progress 13-8

13.5 Paving – Work Not in Progress 13-10

13.6 Seal Coating in Progress 13-12

13.7 Installing and Removing Reflectors 13-14

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PART B – TRAFFIC CONTROL

Section 13: Traffic Control Layouts – Milling, Paving, Seal Coating

LEGEND			
	Flexible Drum		Paint Truck
	Tubular Marker		Escort Truck
	Cone		Chaser Vehicle
	Sign		Vehicle Mounted Rear Crash Attenuator
	Traffic Control Person		
	Work Activity Area		360° Flashing Light
	Work Truck		Portable Traffic Signal
	Shadow Vehicle		Barricade and Fencing
	Shadow Vehicle #1		Flashing Arrow Board (FAB)
	Shadow Vehicle #2		Flashing Arrow Board (FAB) in caution mode
	Buffer Vehicle		Dynamic Message Sign (DMS)

TABLE A – TAPER LENGTHS

Taper Types (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Merge Taper Length	L_M	35	55	160	190	210	230	250	280
Lane Shift Taper Length	L_L	30	50	80	100	110	120	130	140
Downstream Taper Length	L_D	30	30	30	30	30	30	30	30
TCP, Signal, and Shoulder Taper Length (min. 5 devices)	L_S	5	8	15	15	15	15	15	15
Minimum Tangent Length between Tapers	L_T	30	60	160	190	210	230	250	280
Run-In Length on Centreline	L_R	40	50	60	60	70	80	90	100

TABLE B – DEVICE SPACING LENGTHS

Device Spacing (m)		Regular Posted Speed Limit (km/h)							
		≤50	60	70	80	90	100	110	120
Construction Sign Spacing	A	40	60	80	100	150	150	200	200
Buffer Space	B	30	40	60	80	110	140	170	200
Roll-Ahead Buffer Distance	R	30	30	40	40	40	50	50	50
Channelizing Device Spacing for Tapers	C	10	10	15	15	15	15	15	15
Channelizing Device Spacing on Curves and Tangents	D	10	10	30	30	40	40	40	50

13.1 General Information - Milling, Paving, and Seal Coating

Milling, paving, and seal coating are specific types of work that occur on a variety of roadways. The appropriate traffic control layout in this Manual should be chosen for the environment in which the work will occur.

Condition-specific signs should provide drivers with information about the specific site conditions, which may include, but are not limited to:

- low shoulders
- uneven pavement lifts
- no pavement markings
- fresh oil
- loose gravel
- grooved pavement
- dust
- bumps

The signage for low shoulders and uneven pavement lifts requires particular attention because these conditions are not easily detected in darkness or poor weather.

In addition, these principles should be incorporated into traffic control layouts:

1. Warning signs should be repeated as necessary for long sections of affected roadway.
2. Specific signage for motorcyclists and cyclists—for example, the Rough Surface C-019 series and the Bike Hazard C-183 series—should be used in advance of changes to the pavement surface that can affect stability for these road uses, such as gravelled or milled surfaces.
3. The positioning of Construction Speed Zones should be changed as necessary to keep them as short as possible and to avoid requiring drivers to proceed at unreasonably low speeds. Construction Speed Zones should be removed or relocated from areas where the work has been completed so that the speed reductions are specifically appropriate for the active work areas or the areas where hazards exist because of incomplete work.
4. More than one Construction Speed Zone may be used throughout the length of the project, with long zones requiring Maximum Speed R-004 signs and Construction Speed Zone C-080-T tabs to be repeated as necessary. The ends of Construction Speed Zones are to be marked with Maximum Speed R-004 signs that show normal speed limits.
5. For typical pilot car operations, see *Section 4.11.9: Pilot Cars for Work Zones* and *Section 7.16: Pilot Cars.* The signs described in Section 7.16 can also be incorporated into other applications and layouts.
6. Signs should be moved to keep up with moving paving and seal coating operations.

The traffic control layouts are considered the minimum standard. The associated text description of each figure highlights the key standards as well as guidance and options that can be considered by the user. The following information is typical on most layouts in this section:

Standard:

- A No Passing R-022 sign shall be used in areas where passing is prohibited.
- A Passing Permitted R-023 sign shall be used in areas where passing is permitted within the project area.

Guidance:

- None at this time.

Options:

- Where cyclists are regularly observed using the shoulder, a Share the Road W-132-1 sign may be used with an appropriate cycling hazard tab (see *Section 18: Traffic Control Layouts – Bicycle Lanes* for additional information).

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13.2 Benkleman Beam and Falling Weight Deflectometer Testing

The **Benkleman Beam** is a pavement-testing device that measures the deflection of flexible asphalt in order to determine the strength of the road. The worker doing the testing is required to leave the work vehicle to perform the test.

Benkleman Beam testing may take place in travelled lanes. In these cases, a spotter may be beneficial to provide additional situational awareness to the worker regarding vehicle arrivals.

The **Falling Weight Deflectometer** is a pavement-testing device mounted on a small trailer that is towed by a van. Tests are usually made at intervals of between 20 and 100 metres, with the test unit typically stopping for a maximum of 45 seconds per test. The worker doing the testing does not have to leave the work vehicle to perform the test.

Both of these operations are normally classified as mobile work (see *Section 10: Traffic Control Layouts – Mobile Work*).

- For testing on two-lane, two-way roadways with light traffic volumes and good visibility, see *Section 10.3: Intermittently-Moving Work – Two-Lane, Two-Way Roadway.*
- For testing on multilane roadways, see *Section 10.5: Intermittently-Moving Work – Multilane Undivided or Divided Roadway.*
- If traffic volumes are high or conditions prevent traffic from self-regulating, see *Section 7.8: Lane Closure with TCPs – Single Lane Alternating*.
- A buffer vehicle should be used on high-speed, high-volume roadways.
- When Truck Stopped on Road Next 2 km C-038 signs are used for testing operations, the maximum distance between the two opposing C-038 signs should not exceed 2 kilometres except for Falling Weight Deflectometer operations on rural highways, for which it should not exceed 8 kilometres.

13.3 Advance Warning for Paving Work

Purpose:

This layout shows the typical setup for the advance warning area for paving projects.

Paving projects can cover long distances, with work occurring in only one small section at a time, so it is important to identify the project area and the work activity areas separately. Advance warning signage should identify the extent of the project limits and advise road users of upcoming work activity areas.

Standard:

- A dynamic messaging sign (DMS) is required to provide road users with relevant project information.
- A No Passing R-022 sign shall be used in areas where passing is prohibited.
- A Passing Permitted R-023 sign shall be used in areas where passing is permitted within the project area.
- A Paving Ahead Next XX km C-008-1 sign with the appropriate C-008-OL distance overlay shall be used in advance of the Limits of Construction (LoC) to advise drivers of the length of the paving work.

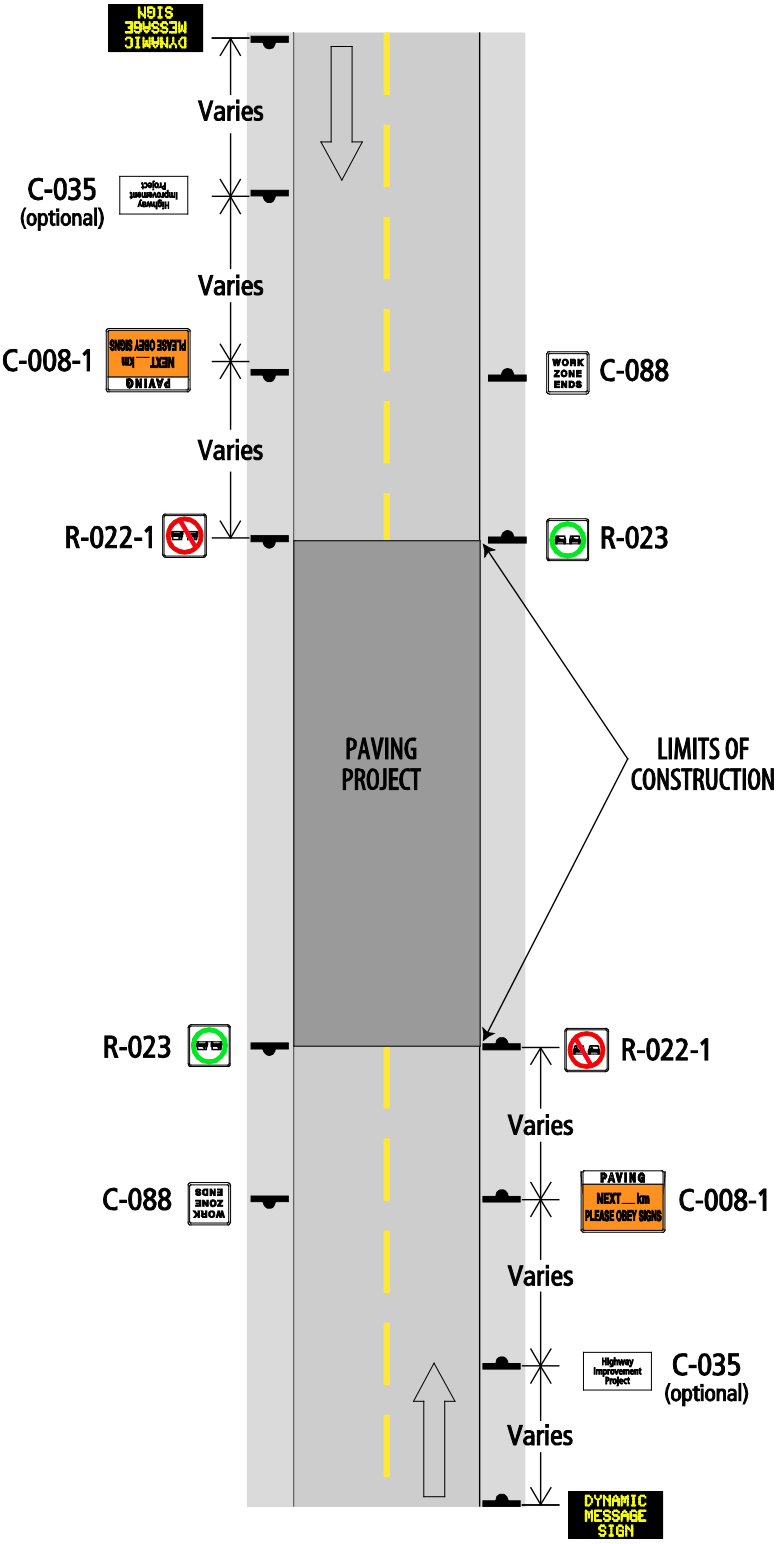
Guidance:

- The dynamic messaging sign (DMS) may be positioned before or after the Construction Project C-035 sign.
- As shown in the diagram, the distance between the Paving Ahead Next XX km C-008-1 sign and the LoC varies as the work activity area moves through the project area.
- When the work is occurring close to the LoC, all other signs preceding the work activity area should be in place first. This may push the C-008-1 sign back from the edge of the LoC during this period.
- See *Section 13.4: Paving – Work in Progress* for advance warning signage requirements for a work activity area that is located close to the LoC.
- Paving Ahead Next XX km C-008-1 signs with the appropriate C-008-OL distance overlays should be repeated at various distance intervals in advance of the active work area.

Options:

- Additional signs may be required throughout the advance warning area and repeated as necessary (see *Appendix B: Standard Construction Signs*). These may include, but are not limited to:
  - Soft Shoulder C-012
  - Fresh Oil C-014
  - Single Lane Traffic C-030-8
  - Pavement Ends C-149

Figure 13.3: Advance Warning for Paving Work



13.4 Paving – Work in Progress

Purpose:

This layout shows the work activity area within the longer project area (Limits of Construction). It is important to focus on the work activity area and the current conditions to ensure that the most applicable signage and devices are used and that the selected layout accurately reflects the work.

Standard:

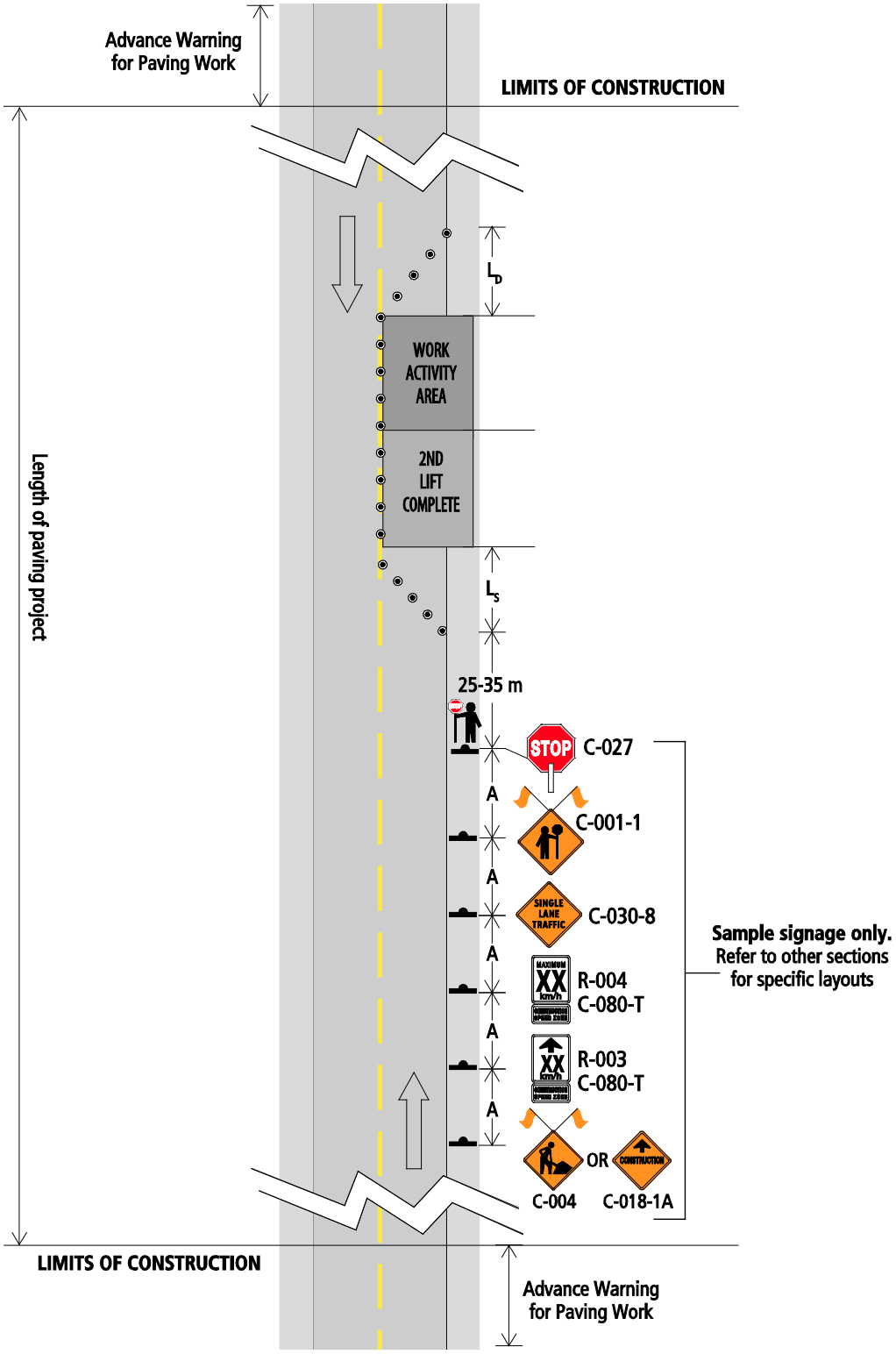
The most appropriate layout should be applied to the work activity area, based on the highway type and traffic control required. The layout options include, but are not limited to, those listed below.

Type of Work Area	Traffic Control Layouts
Two-Lane, Two-Way Single Lane Alternating	*Section 7.8 Lane Closure with TCPs – Single Lane Alternating*
Pilot Car Operation	*Section 7.16 Pilot Cars* *Section 13.6 Seal Coating in Progress*
Multilane Roadway Lane Closure	*Section 8.6 Right Lane Closed* *Section 8.7 Left Lane Closed* *Section 8.8 Centre Lane Closure (< 60 km/h)* *Section 9.7 Left Lane Closed* *Section 9.9 Centre Lane Closure (≥ 70 km/h)* *Section 9.10 Double Lane Closure*
Lane Shift	*Section 8.10 Centreline Crossover* *Section 8.14 Passing/Climbing Lanes – Lane Shift* *Section 9.11 Median Crossover*
Intersection	*Section 11.9 Left/Right Lane Closure within Intersection – Multilane Intersection* *Section 11.10 Right Lane Closure with Right-Turn Lane (Near Side) – Channelized Right Turn Open – Multilane Intersection* *Section 11.11 Two Lanes Closed (Near Side) – Multilane Intersection* *Section 11.12 Two-Lane Closure – Multilane Intersection*

Guidance and Options:

- The Guidance and Options for the selected layout apply.

Figure 13.4: Paving – Work in Progress



13.5 Paving – Work Not in Progress

Purpose:

This layout shows a typical setup for a paving project that is under way—but for which workers are not currently present (e.g., at night or on a weekend)—and that has site conditions of which drivers should be advised through appropriate signage (e.g., No Centreline, Bump, Low or Soft Shoulder, No Passing, etc.) These signs may also be used when paving is in progress.

Standard:

Signs commonly required on inactive or dormant paving projects include, but are not limited to:

- Construction Ahead C-018-1A sign with Construction Speed Zone C-080-T tab for long-duration work.
- Paving Next XX km C-008-1 signs in advance of and throughout the work activity area at 5-kilometre intervals.
- Low Shoulder on Left/Right C-013-LR sign where shoulder is lower than road surface.
- Uneven Pavement on Left/Right C-010-LR sign where there is uneven pavement on either side of the travel lane (excluding shoulders).
- Bump or Rough Roadway Ahead C-017 sign where sharp road surface changes are sufficiently abrupt.
- Uneven Pavement Ends C-016 sign at the end of an uneven section of pavement.

Other signs that help to identify certain conditions or activities on dormant sites include:

- Soft Shoulder C-012
- Loose Gravel C-015
- Pavement Ends C-149
- Use Headlights – Extreme Dust C-185-3

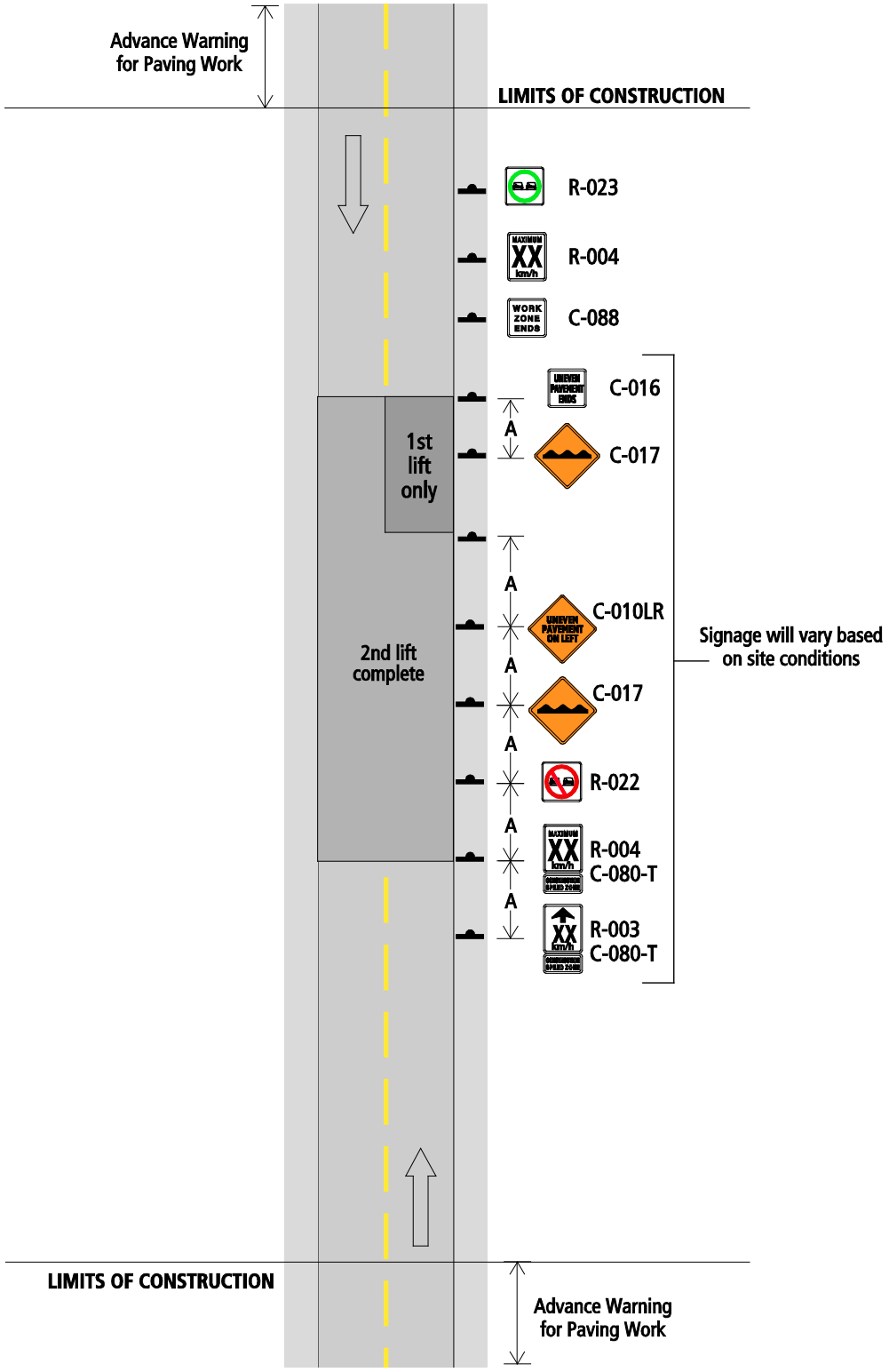
Guidance:

- When the roadway and shoulders are clear of machinery and obstructions, and the condition of the unfinished roadway is such that traffic can proceed safely without the assistance of TCPs or a pilot car:
  - All non-applicable signs should be removed or covered.
  - The Paving Ahead Next XX km C-008-1 sign with the appropriate C-008-OL distance overlay should be repeated every 5 kilometers, showing a decreasing distance to the end of the project.

Options:

- Additional signage may be required throughout the work area and the project area, and repeated as necessary (see *Appendix B: Standard Construction Signs*).

Figure 13.5: Paving – Work Not in Progress



13.6 Seal Coating in Progress

Purpose:

This layout shows the typical setup for an active seal coating project on a two-lane, two-way roadway. Seal coating projects can involve a long work activity area so a pilot car operation is often used with a single lane alternating setup to guide traffic through the work zone.

Standard:

- A Seal Coating – Loose Gravel Next XX km C-008-2 shall be positioned in advance of the work activity area.
- If TCPs are used, a Traffic Control Person Ahead C-001-1 sign is required.
- When TCPs are directing traffic, the construction speed limit shall be ≤ 70 km/h.
- Overhead lighting shall illuminate each TCP location at night.
- A Follow Pilot Car C-049 shall be positioned on the shoulder.
- The double-sided Pilot Car (frontward-facing) and Pilot Car Do Not Pass (rearward-facing) C-048-1-DS sign or the Pilot Car C-048-2 overhead sign are required on the pilot car so that its signage is visible from both directions.
- A Work Zone Ends C-088 sign, followed by a Maximum Speed R-004 sign, shall be positioned the end of the work activity area.
- The Seal Coating – Loose Gravel Next X km C-008-2 sign should be repeated every 5 kilometers, showing a decreasing distance to the end of the project.

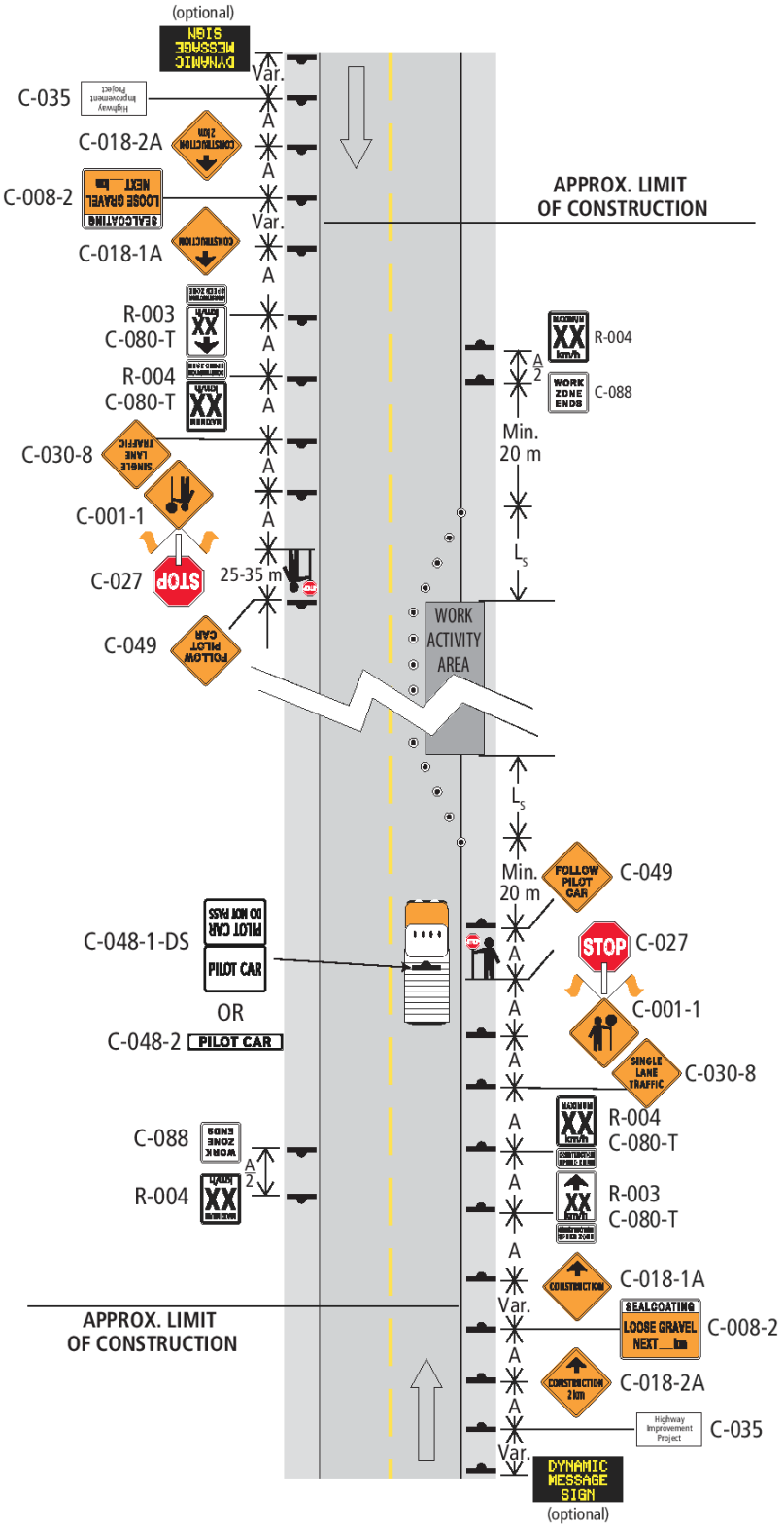
Guidance:

- The pilot car operator and TCPs should be in radio contact.

Options:

- The Flagger Ahead C-001-2 sign or Prepare to Stop C-029 sign may be used for additional advance warning where TCPs are stopping traffic.
- Additional signage may be required throughout the advance warning area and repeated as necessary (see *Appendix B: Standard Construction Signs*), including but not limited to:
  - Soft Shoulder C-012
  - Fresh Oil C-014
  - Loose Gravel C-015
  - Single Lane Traffic C-030-8
  - Follow Pilot Car C-049
  - Sweeper Working C-076
  - Pavement Ends C-149
  - Truck Crossing/Entering Highway C-172-R
  - Use Headlights – Extreme Dust C-185-3

Figure 13.6: Seal Coating in Progress



13.7 Installing and Removing Reflectors

Purpose:

This layout shows two potential setups—depending on the shoulder width—for the installation of temporary and permanent reflectors on a hard surface and the removal of reflectors from such a surface:

- *Figure 13.7 A – Insufficient Shoulder Width*
- *Figure 13.7 B – Sufficient Shoulder Width*

These setups are appropriate for intermittently-moving, short-duration work during daylight hours on a low-speed, low-volume, two-lane, two-way roadway.

Night work and roadways with higher speeds and volumes require more complex setups. Refer to these sections for appropriate traffic control layouts:

- *Section 7: Traffic Control Layouts – Two-Lane, Two-Way Roadways*
- *Section 8: Traffic Control Layouts – Multilane Undivided Roadways*
- *Section 9: Traffic Control Layouts – Multilane Divided Roadways*
- *Section 10: Traffic Control Layouts – Mobile Work*

Standard:

For areas with **insufficient shoulder width:**

- A shadow vehicle shall be used to protect workers and provide advance warning to approaching motorists.
- The work and shadow vehicles shall both display a rear-mounted Slow Moving Vehicle C-036 sign, a flashing arrow board (FAB) in bar mode, and 360-degree flashing lights and 4-way flashers.
- Traffic shall be regulated with a single lane alternating setup, either by using a Yield to Oncoming Traffic R-056-1 sign on the back of the shadow vehicle or by using Traffic Control Persons.
- A Slow Vehicle Next XX km C-044 sign or a Truck Stopped on Road Next 2 km C-038 sign shall be used in advance of the work and identified with flags. It shall be moved as the work progresses so that the work vehicle remains within the distance indicated on the sign.
- If 5 or more vehicles are queued or if potential passing hazards are of concern, the work and shadow vehicles shall move off the road to allow traffic to pass.
- If passing traffic is an ongoing concern, other traffic control methods and layouts shall be used.
- The Slow Vehicle Next X km C-044 sign shall be moved as the work progresses so that the work vehicle remains within the distance indicated on the sign.

For areas with **sufficient shoulder width:**

- The work and shadow vehicles shall both display a rear-mounted Slow Moving Vehicle C-036 sign, a flashing arrow board (FAB) in bar mode, and 360-degree flashing lights and 4-way flashers.
- Traffic may be permitted to pass on the shoulder if a 3.5-metre distance can be maintained between the work/shadow vehicle and the edge of the paved shoulder.
- If traffic is permitted to pass on the shoulder side, a Pass This Side C-042-R (right arrow) sign shall be displayed on the rear of the shadow vehicle.
- A Slow Vehicle Next XX km C-044 sign shall be used in advance of the work and identified with flags. This sign shall be moved as the work progresses so that the work vehicle remains within the distance indicated on the signs.

Guidance:

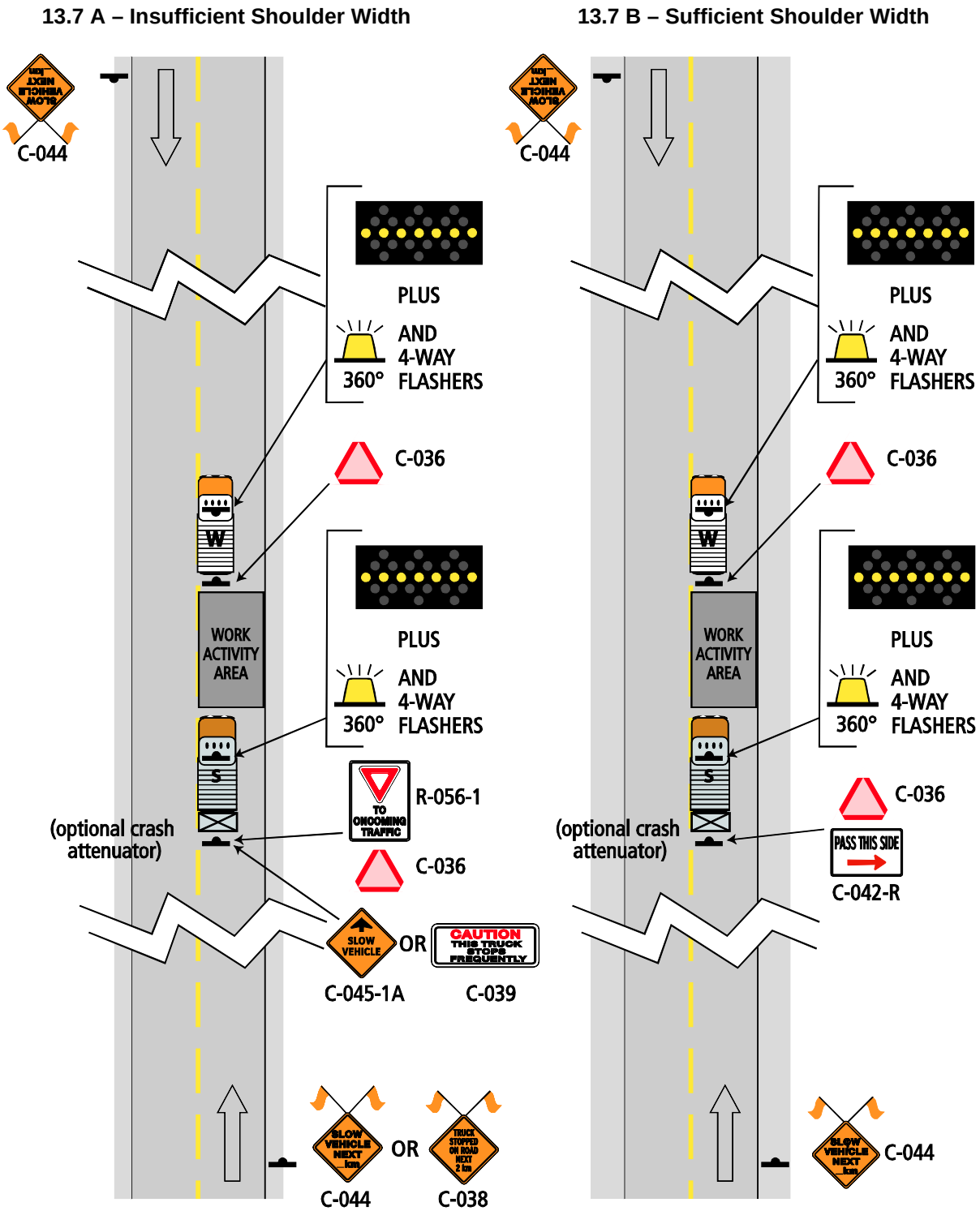
- The distance shown on Slow Vehicle Next X km C-044 signs should not exceed 8 kilometres.
- All temporary signs should be removed or covered when work is not in progress.
- Work vehicles may travel at posted speeds when work is not in progress.

Options:

- The Road Authority may require that the shadow vehicle have a rear-mounted Slow Vehicle(s) Ahead C-045 sign or other appropriate sign.
- A dynamic message sign (DMS) displaying the appropriate directional text and arrow may be used instead of the Pass This Side C-042-R sign.

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Figure 13.7: Installing and Removing Reflectors



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